

Punnam Yuva Keerthana Venkata Gowrish

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EDUCATION

Indian Institute of Technology Bhubaneswar

8.98/10

B.Tech in Electronics and Communication Engineering

Aug 2024 – May 2028

Sri Chaitanya Junior College – Intermediate (12th)

98.1%

Board of Intermediate Education

Jun 2022 – Jun 2024

Sri Chaitanya Techno School – Secondary School (10th)

572/600

Board of Secondary Education, Telangana

Jun 2021 – May 2022

TECHNICAL SKILLS

HDL/RTL: Verilog HDL, SystemVerilog (Basics), RTL Design, RTL Verification, FSM Design, Digital Logic Design

ASIC and VLSI: CMOS Fundamentals, RTL-to-GDSII Flow, Logic Synthesis, Static Timing Analysis (STA),

Floorplanning, Placement, Clock Tree Synthesis (CTS), Routing, Physical Design

Digital systems: RISC-V (RV32I), Computer architecture, Pipelined Processor Design, UART, Floating-Point Arithmetic, Communication Protocols, FPGA Design

EDA Tools: Vivado, Yosys, OpenSTA, GTKWave, ModelSim, VS Code, Git

Programming: C, C++, Python, RISC-V Assembly, TCL (Basics)

PROJECTS

RV32I RISC-V Processor Family | *RTL Design Project* | **GitHub Repository**

Jan 2026 – Mar 2026

- Designed two 32-bit RISC-V (RV32I) processors in Verilog HDL — a single-cycle architecture and a 5-stage pipelined architecture (IF, ID, EX, MEM, WB) — supporting 35+ RV32I instructions.
- Enforced pipeline registers, data forwarding, and a hazard detection unit in the pipelined design to resolve data and control hazards and improve instruction-level parallelism.
- Executed a comprehensive testbench suite covering 40+ instruction sequences; validated functional correctness against the official RISC-V compliance test suite with zero mismatches. Deployed on Xilinx Basys3 FPGA via Vivado.

RTL-to-GDSII Flow for Synchronous FIFO | *ASIC Design Flow* | **GitHub Repository**

June 2026 – Aug 2026

- Engineered a parameterized Synchronous FIFO in Verilog HDL, implementing read/write pointers, memory architecture, and full/empty flag generation with a comprehensive functional testbench.
- Executed the complete ASIC RTL-to-GDSII flow using Yosys, OpenSTA, and OpenROAD, including logic synthesis, Static Timing Analysis (STA), floorplanning, power planning, placement, Clock Tree Synthesis (CTS), and routing with the Nangate45 standard cell library.
- Developed TCL automation scripts and applied SDC timing constraints to analyze and optimize timing (WNS/TNS), area, power, and critical paths, generating a synthesized netlist and routed DEF layout.

BFloat16 Floating-Point Arithmetic Unit | *Digital Design Project* | **GitHub Repository**

May 2026 – Jun 2026

- Developed a 16-bit BFloat16 floating-point arithmetic unit in Verilog with a 2-bit operation-select input with correct rounding and special case handling (NaN, Inf, zero).
- Structured the design into seven RTL stages to handle mantissa alignment, computation, normalization, and result repacking. Covered edge cases, validating output against expected results across multiple test vectors in simulation.
- Synthesized the processor onto FPGA hardware with LED-based output monitoring for real-time result verification.

UART Communication System | *Communication Protocol Project* | **GitHub Repository**

Oct 2025 – Dec 2025

- Architected a complete UART communication system modular RTL architecture, implementing the standard 8N1 frame format (8 data bits, no parity, 1 stop bit) at 9600 baud, with FSM-based transmitter and receiver modules.
- Implemented two data-flow configurations: a normal mode following TX → RX → process → dataout, and a bidirectional mode following TX → RX → process → TX again → dataout, where received data is processed and re-transmitted to confirm round-trip communication integrity.
- Built a dedicated baud rate generator module to derive accurate 9600 baud bit timing from the Basys3's 100 MHz onboard clock, using separate TX/RX baud tick generation.

ACHIEVEMENTS

- Successfully solved all HDLBits exercises, demonstrating proficiency in Verilog HDL, RTL Design, Digital Logic Design, FSM Design, and Hardware Modeling.